Quiz 3

Expectation and retrieval

Try each question before reading the worked solutions on page 2.

1. $P(X = 0) = 1/4$ and $P(X = 4) = 3/4$. What is $\mathbb{E}[X]$?

A. 1

B. 2

C. 3

D. 4

2. I_A indicates an event with $P(A) = 0.3$. What is $\mathbb{E}[I_A]$?

A. 0.3

B. 0.7

C. 1

D. 0

3. $\mathbb{E}[X] = 2$ and $\mathbb{E}[Y] = 5$. Without assuming independence, what is $\mathbb{E}[3X - Y]$?

A. 11

B. -3

C. Cannot be determined.

D. 1

4. N counts successes in 12 independent Bernoulli trials with success probability $1/4$. What is $\mathbb{E}[N]$?

A. 12

B. 3

C. $1/4$

D. 9

5. A PMF has support $\{1, 2, 4\}$. Which identity always gives $P(1 < X \leq 4)$?

A. $F_X(4) - F_X(1)$

B. $F_X(4) - F_X(2)$

C. $F_X(4)$

D. $F_X(1)$

Worked solutions

1. Expectation as a weighted average

Answer C

Multiply each possible value by its probability, then add:

$$\mathbb{E}[X] = \sum_{x \in \{0,4\}} xP(X = x) = 0\left(\frac{1}{4}\right) + 4\left(\frac{3}{4}\right) = 3.$$

The probabilities are the weights. An expectation need not itself be a possible value of X.

2. The mean of an indicator

Answer A

The indicator equals 1 when A occurs and 0 otherwise. Therefore

$$\mathbb{E}[I_A] = 1 \cdot P(A) + 0 \cdot P(A^c) = 1(0.3) + 0(0.7) = 0.3.$$

In general, the mean of an event indicator equals the probability of that event.

3. Linearity without independence

Answer D

Linearity allows constants to be taken outside expectation and sums or differences to be evaluated term by term:

$$\mathbb{E}[3X - Y] = 3\mathbb{E}[X] - \mathbb{E}[Y] = 3(2) - 5 = 1.$$

This property does not require independence; the two given finite means are sufficient.

4. Counting successes using indicators

Answer B

Let I_i indicate success on trial i. Then $N = \sum_{i=1}^{12} I_i$ and every indicator has mean $1/4$. By linearity,

$$\mathbb{E}[N] = \sum_{i=1}^{12} \mathbb{E}[I_i] = 12\left(\frac{1}{4}\right) = 3.$$

This is the binomial mean np. Independence is given, although summing these expectations does not require it.

5. CDF subtraction and endpoints

Answer A

$F_X(4)$ includes the masses at $1, 2, 4$. Subtracting $F_X(1)$ removes the mass at 1, leaving 2 and 4:

$$P(1 < X \leq 4) = P(X \leq 4) - P(X \leq 1) = F_X(4) - F_X(1).$$

In general, $P(a < X \leq b) = F_X(b) - F_X(a)$ for $a < b$.